Grind & Crumb Hackathon Report - Group 2 (Hugo Sharpe, Anas Mohammed, Peter Galambos, Temi Dada)

Introduction

This proposal outlines strategic recommendations for Grind & Crumb (G&C) to improve sustainability and human impact, and the allocation strategy of the funding including the relocation of the Manchester site for £250,000 in annual rent savings, automate coffee prep, repair equipment, improve app infrastructure, and introduce a promotional campaign to encourage app usage.

Analysis using the tool

We built a modular validator to inspect Grind & Crumb's 22 spreadsheets across 45 automated checks and corrections before analysing the data, revealing 5 warnings and 10 critical failures - including hundreds of duplicate customer and transaction IDs (Fig. 1).

Through review and survey analysis within our tool utilising the Gemini 2.5 Flash model to identify customer sentiment, we identified key issues facing various branches. There is a sentiment that the stores in Manchester are understaffed especially during the lunch rush hour. Additionally, missing food in orders and food temperature/condition were consistently complained about across all stores (Fig. 2)

It was also noted that 11 of 12 stores had user complaints regarding app checkout crashes and vanishing stamps. This was reflected in the monthly P&L graph in the tool showing a decrease in net sales and net profit across all stores following the v3.1 app hotfix attempt. Alongside this, we can see that the ‘App Click & Collect’ over the year made up 12.2% of sales, with sale volume significantly decreasing to 9.3% for April ‘26 following the v3.1 update. This suggests that a significant app update is required alongside a promotional program to encourage customers back to the app following their bad experiences.

Through our tool’s Huff Gravity Cannibalisation Model, users of the tool can input csv files for additional candidate sites and determine the viable net profit, accounting for cannibalisation from existing sites and the cost of fitting. From the given candidate_sites.csv, our model identified Leeds Headingley as the prime candidate for expansion (Fig. 3).

Our tool also analysed the sales channels and identified the third-party delivery as particularly unprofitable with a net margin of -21.4%. This was due to the large platform commission, as well as the extra labour cost for packaging (Fig. 4).

Proposals for the £1.5m investment

To deal with the understaffing of Manchester branches, we propose that we replace the unprofitable Manchester Piccadilly site with Manchester Oxford Road with fitting costs repaid within a year from rent savings alone [1]. The refitting would be less expensive than

projected if equipment was repurposed. Labour costs could be reduced by automating the food/drink process through machines like the Eversys ShotMaster Pro, this would also help combat understaffing during peak times. We propose the acquisition of 6 machines from Bridge Coffee Roasters [2]. If the trial run was successful, this could be implemented in more branches, allowing for potential labour cost reductions by reducing the number of baristas.

To combat complaints with food temperature and conditions, we propose G&C invest in maintenance for all machines (coffee,fridge,heating) across all stores. [3]

To increase the usage of the click and collect app, G&C should invest in quality assurance, testing and maintenance of the app. This would increase customer trust in the brand, driving greater sales [4]. To increase awareness of the app, an advertising promotion campaign should be run to capture new audiences, and to make existing customers aware of the benefits of the app [5].

We also recommend G&C to invest in an expansion into Leeds Headingley, with a net profit of £128,000 per annum, with building costs being recouped within 30 months. This site offers the greatest ROI using our model [6].

To mitigate the money lost from delivery platform commissions, G&C is encouraged to increase costs of items on delivery platforms by a minimum of 25%. [7]

The total costs of all these changes comes to £863,220, allowing the remaining money to be allocated to any unexpected expenses from opening new stores, as well as continuing the promotional advertising campaign if deemed necessary. This money could also be committed to the implementation of charity and sustainability initiatives as outlined in the following section [8].

Charity and Sustainability Initiatives Recommendations

We recommend an increase in G&C's positive impact on the environment and communities through a new scheme that encourages customers to switch to using their own reusable cups by increasing the cost of hot drinks by £0.25 if a reusable cup is not used. Research conducted by EPECOM on behalf of the Scottish Government found that additional surcharges were more effective at influencing customer behaviour than small benefits (Poortinga et al., 2019). This additional money would be donated to The Fairtrade Foundation alongside the option on card terminals for customers to round-up their bill to the nearest pound to charity. The Fairtrade Foundation works to improve the lives of coffee farmers across the world through providing a safety net against volatile coffee prices, as well as investment in climate change resistant crops (*The Coffee Industry*, n.d.). We propose a round-up as it is shown by Hwang that round-ups are more effective than flat donation requests in crowded environments such as a coffee shop (Hwang et al., 2021).

Upon analysing the waste data in the tool, we found that on average ~£11,000 is wasted per store annually. We suggest that the stores could sign up to Too Good to Go to reduce CO2 emissions and food waste, whilst also recovering some of the losses(Sgroi et al., 2024). Assuming that each store sells 5 bags for £5 per bag each store could recover up to £9000 in revenue annually. The remaining food waste could be donated to food shelters for free.

Appendix

Figures

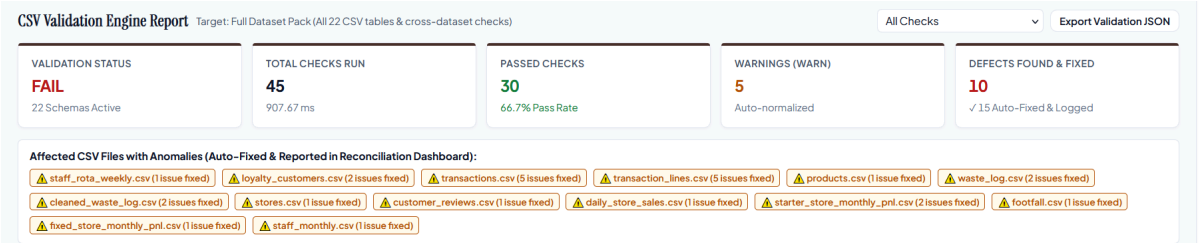


Fig 1. Validation checks and results.

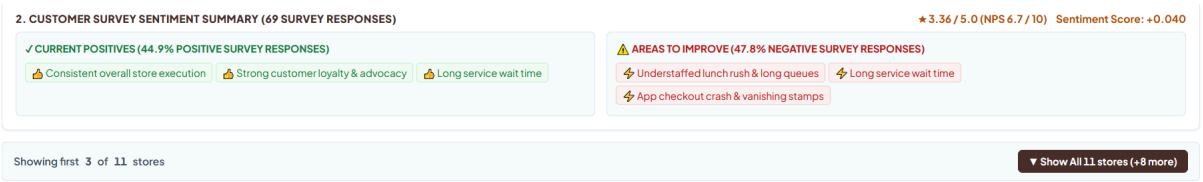


Fig 2. Reviews complaining about app checkout crash and vanishing stamps across 11/12 stores, alongside understaffed lunch hours.

Site ID	Candidate Site	City & Type
CS1	Leeds Headingley	Leeds (High street)

Est. Avg Spend	Est. Revenue/mo	Est. Costs/mo	Standalone Profit	Cannibalized Profit	Affected Stores (Huff Pull)	Net Chain Profit/mo	Fitout & Payback	Strategic Verdict
£6.09	£60,522.42	£49,653.04	£10,869.38 (18% margin)	-£213.84 (-83 cust/mo)	S03 (2.5mi: -0.4%) S12 (2.65mi: -0.4%)	+£10,655.54 £127.9k/yr	£320.0k 30 mos (40% ROI)	★ PRIME EXPANSION

Fig 3. Huff Gravity model determining net profit and payback period.

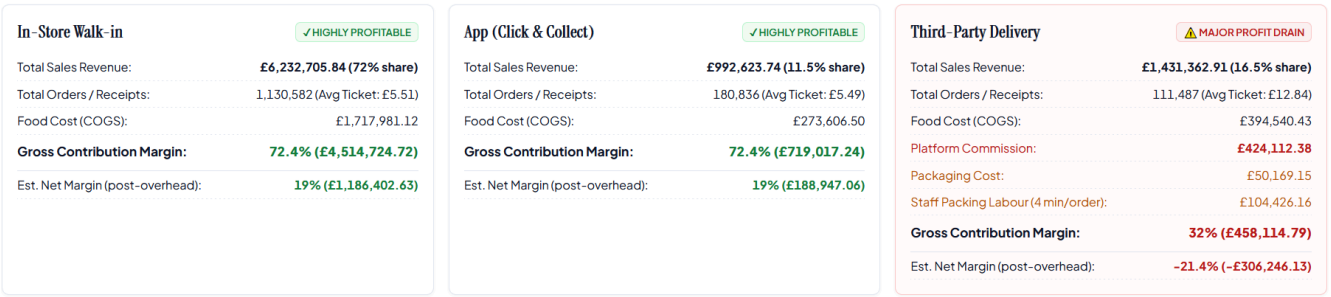


Fig 4. Overall margin for each sales channel and contributions to cost, with 3rd party being least profitable.

Calcs

[1] (£30500 - £9800 ~= £21k/m, £248k/yr)
[2] 6*42,910 = £257,460 for renting 2 machines per branch in Manchester
[3] £14760 for maintenance of coffee machines, fridges/cooling appliances and heating appliances.
[3.1] Coffee machines on average cost £40 to diagnose and £210 to repair per (The Coffee Repair Clinic, 2026).
[3.2]Assuming 2-3 coffee machines per site, cost maximum £630 per site for full diagnostics + repair. For 12 sites sum to £7560.

[3.3]According to (Ascot Wholesale, 2026) maintenance on fridges is ~£400 yearly with ovens costing £300 - since we are using heaters instead of ovens presumably cheaper, so £200/yr for £600 per site
[3.4]Total maintenance costs: £14,760/yr for all sites
[4] 15,000 for QA + fixing + testing and maintenance of the app
[5] Advertising campaigns to capture new audiences ~£15,000/month (teamwork.com, 2026)
[6] £320.0k CS1 with potential profit of £10,869/month (30 months to recoup costs)
[7] Platform commission/orders placed = 424,112.38/111,487 = £3.80 lost per order (loss/order) / avg ticket spend = 3.80 / 12.84 * 100 = 29.6% price increase to offset commission
[8] Running total: £863,220 - rest can be saved as liquid assets for use in emergencies.

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